Network Protocols

Protocol 1 – TCP

TCP protocol is relevant for the Global Dimensions College since a large amount of data is transmitted between computers, servers, and other devices on the network. This protocol prevents research papers, emails, and other important documents from being corrupted. On the other hand, academic staff and administrators are responsible for the student's information and other management systems wherein data transactions are frequent. Most importantly, it enables collaboration internationally as the TCP ensures that the data packets that are sent over long distances won't be lost as well as other academic activities.

Protocol 2 – DHCP

In the Global Dimensions College, since numerous devices will be connected in a network, it would be efficient for the DHCP protocol to automate the allocation of IP addresses to each device, particularly on laptops and phones instead of the administrator doing it. Even when the students, and staff move from different locations or disconnect, assigning new IP addresses would be easier if connected to another network segment.

Protocol 3 –DNS

This is relevant in the Global Dimensions College scenario since resources can be accessed within a network such as from the UK to Manila with the domain names. The students and staff can access school websites, applications, and services and send emails to each other readily. This will be also useful for locating resources, and servers, in a network.

Analysis

Based on the scenario of the Global Dimensions College, the most suitable topology for the networking plan is the star topology. Star topology is very reliable since for example, if one cable from a computer fails, the other computers will still work (GeeksforGeeks, 2022). It is easier to set up and less expensive since it only requires a port, switches, and one router, making it ideal for a university environment. Data collisions will be prevented as the data travels through a central point wherein it can be easily monitored by the IT department in case of maintenance.

Tree topology can also be applied to specific structures of the network plan. Given that the networked devices are connected to the switch, these switches are connected to the main switch and then to the campus router, which illustrates tree topology or hierarchical structure. Its like the main campus switch is the root while the sub-switches and computers are the branches.

Commonly, school computers in a computer lab or office share the same model or brand, and most likely, they have the same number of ports. This implements star topology ideal for the school environment since computers can be easily connected to the switch with an Ethernet cable. Additionally, connecting new computers to the network would be easier as it will only need a port. Star topology is commonly used in schools and offices since it is the most dependable and cost-effective topology. (Heimduo, 2020)

The star topology will be implemented by connecting the computers from each room to a switch, wherein each switch will be connected to the main campus switch. The main switch is connected to the router and this handles the communication between the devices on different room networks and the Internet.

Moreover, since LAN is the communication network of a university, star topology is said to be the most widely used topology for LANs. The scenario also implements a WAN network as the Philippines campus will be connected to the UK campus. The routers must have LAN and WAN connections which will enable communication between both campuses. This will be done by having a serial connection between routers and static routing.

Servers for Global Dimensions College

Server 1 - Database server

Since the scenario is a university setup, it is essential to implement a Database Server for the administrators to access, retrieve, or update data related to students, faculty, or staff information. Along with operations encompassing financial management, and human resources, and other relevant information. This centralised database is the place for making informed decisions and ensuring the security of university information.

In terms of cost-effectiveness, it is practical and applicable since the database management system is free to download such as MySQL 8.0. It is the updated version of the software and can deal with large amounts of data. According to the official website of MySQL, the performance and security features are enhanced in this version which can be advantageous for the university administrators and IT team (Oracle, 2000). This version can support long-term use and is scalable for growing data in a university environment.

Server 2 - File Server

It is also essential to implement a File Server for the faculty and administrators to access important documents and collaborate efficiently without the need to struggle to transfer files from one computer to another each time. Course-related materials can be easily accessed by different faculties and can serve as a data backup for various documents. If necessary, students can also submit their assignments digitally in a dedicated directory on the file server by the university network.

According to the official website of File Cloud, it is a file server that is frequently used by universities as it offers secured storage, file sharing, collaboration, unlimited account access, and offline access (File Cloud, n.d.). In terms of cost-effectiveness, it removes the need for setting up hardware materials or physical servers. In that case, it allows remote access, which is imperative for both campuses of Global Dimensions College. The performance is seamless and supports all major operating systems, ensuring compatibility for all users.

Server 3 - Proxy Server

Since the aim is to provide direct internet access from the Brighton, UK campus of Global Dimension College, it is necessary to implement a proxy server to tighten the security of websites that can potentially threaten the private network of the university. The proxy server will ensure a

safe browsing environment for different users in a university by blocking malicious content and preventing unauthorized access.

With regards to cost-effectiveness, it can save bandwidth usage since it compresses the web traffic accessed by multiple users and takes down the ads from websites. Thus, reduces the cost related to the data consumption from the Internet Service Provider. A popular proxy server and manager is Bright Data, wherein is used by known universities such as Princeton and Oxford University with a reasonable price for its numerous proxy features. (Bright Data, n.d.).

Server 4 - Web Server

Web Server is important to implement because the GDC official website is hosted by a web server, thus serving as a primary platform for the faculty, students, and public wherein they can access information about the university, events, admissions, and programs. Not only for the dissemination of information, but also for the management system of students' portals and learning platforms such as access to assignments, courses, and grades.

Surely there are open-source web servers that are free to use and also paid versions for additional features. For example, Apache Server is a recommendable software for web servers as it is optimal and highly flexible for various types of websites.

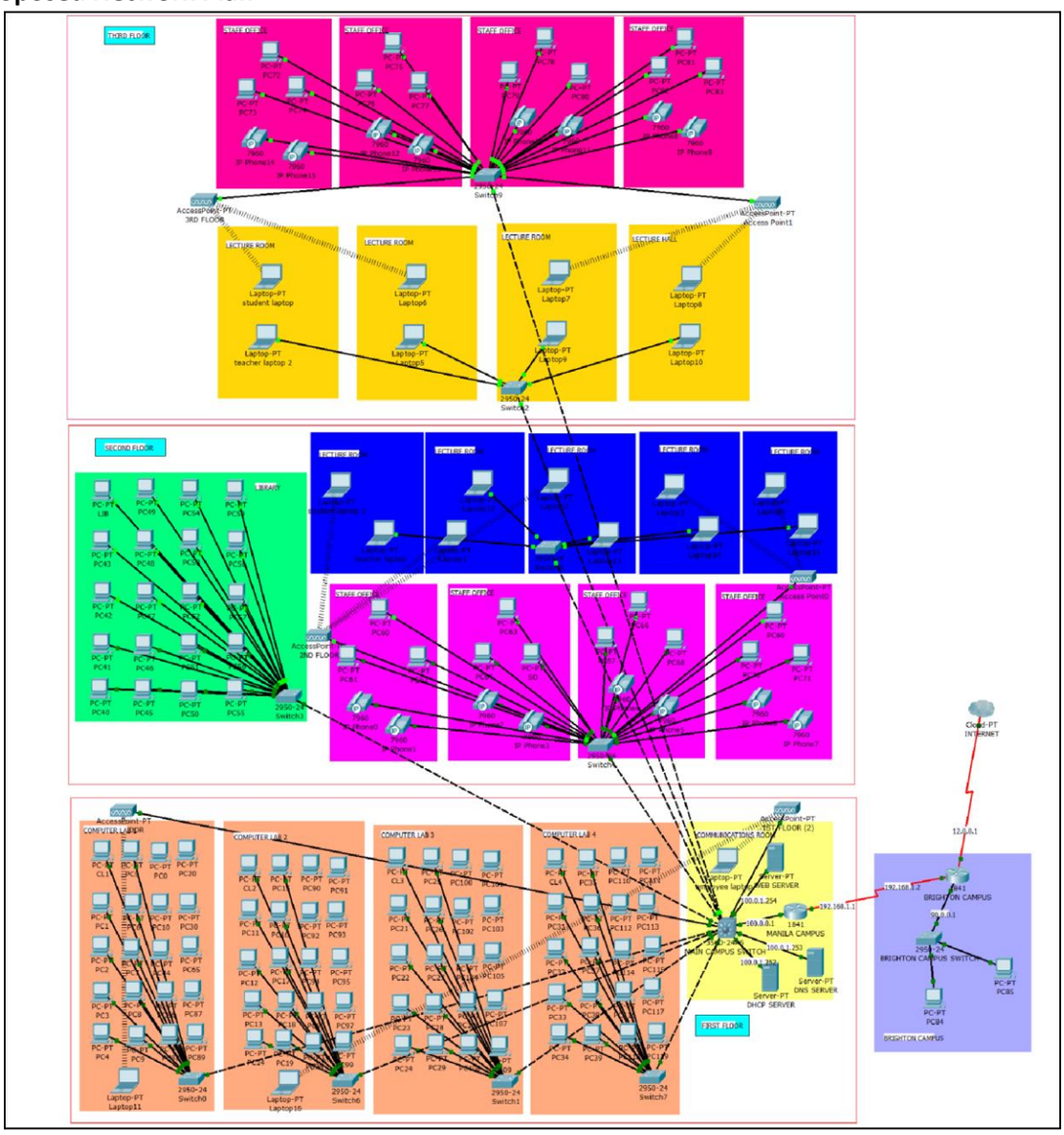
Server 5 - DHCP Server

DHCP Server simplifies the workload of the IT team in the university and reduces the risk of errors for manual configuration. Since numerous laptops and phones are continuously connecting and disconnecting to the network, centralized IP management or the DHCP server would be efficient in allocating IP addresses in a university. There are DHCP server software available and commonly free to use, like Windows DHCP Server. This server is ideal for dynamic environments as it improves efficiency for network access of users.

Server 6 - DNS Server

DNS Server is crucial for naming university websites as well as easing the navigation and retrieval of information. It also facilitates domain names for email services which play a huge part in the university environment, whether it be faculty, staff, or students. Examples of DNS servers are BIND (Berkeley Internet Name Domain), Microsoft DNS, Google, and more. Most of these DNS servers are free to use and won’t be a problem implementing it.

Proposed Network Plan



The main network topology that I implemented for GDC is a star topology, supplemented by the integration of a tree topology. Thus, can be classified as Hybrid Topology. My network diagram illustrates that computers and servers are connected to respective switches near the devices' positions, establishing a star topology.

It is classified as a tree topology due to the interconnection of the sub-switches or room switches to the Main Campus Switch, and this switch is linked to the Manila Campus Router. Additionally, the laptops are wirelessly connected to the designated access point on each floor, and these access points are connected to the room switches, forming a tree topology. My network diagram is divided into rectangular sections representing the floors, and each room is distinguished by colors.

It also illustrates in my network plan that each floor has 2 access points instead of just one. This is because in real-life scenarios, having 2 access points in a network is a common practice. Since this will implement redundancy to minimize the downtime (Ariola, 2023). The APs are also installed in the hallway so that it covers each area of the rooms.

The cloud symbol or the Internet provided by the Brighton campus has an IP address of 12.0.0.1 and I added static routing configuration for both routers, which enables internet access to Manila Campus.

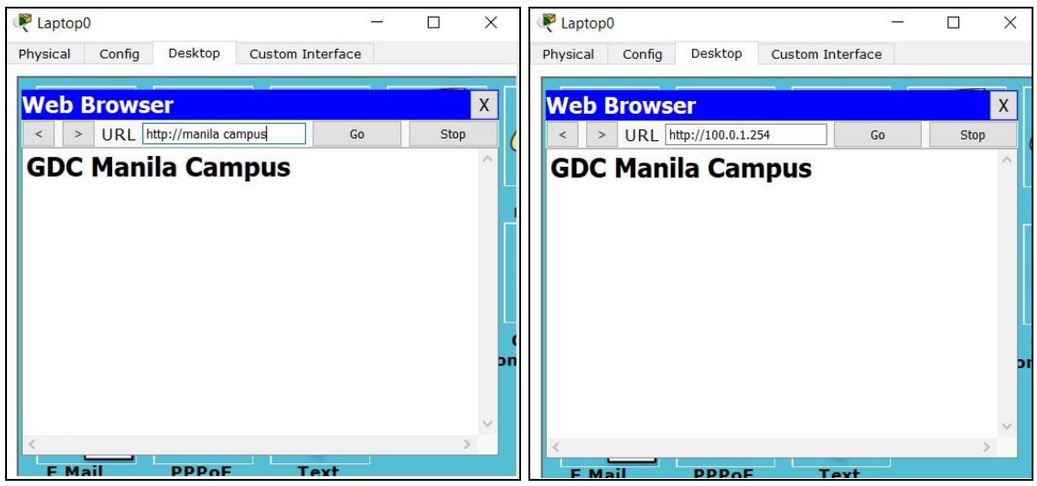
Brighton Campus

- On the right side of the network plan, the Brighton Campus network can be seen. Starting with the Brighton Campus router, since they will be providing the internet connection, I connected the cloud to the router to represent the internet. The Brighton Campus can provide internet access to the Manila Campus by facilitating point-to-point serial link or communication with the use of serial cable, which creates a WAN infrastructure. The serial connection was made possible by connecting a HWIC-2T module in the router.
- The IP address I assigned for the LAN interface of the Brighton Campus Router is 90.0.0.1 and the network address will be 90.0.0.0, making it the gateway of the PCs inside the network.
- The IP address I assigned for the WAN serial interface is 192.168.1.2 which is a component for static routing of the two routers. This configuration allows the routers to forward packets both ways.
- It can also be seen that the serial interface IP address of the Manila Campus Router, 192.168.1.1, is in the same subnet as the Brighton Campus Router to ensure that they can communicate properly and accurately route.

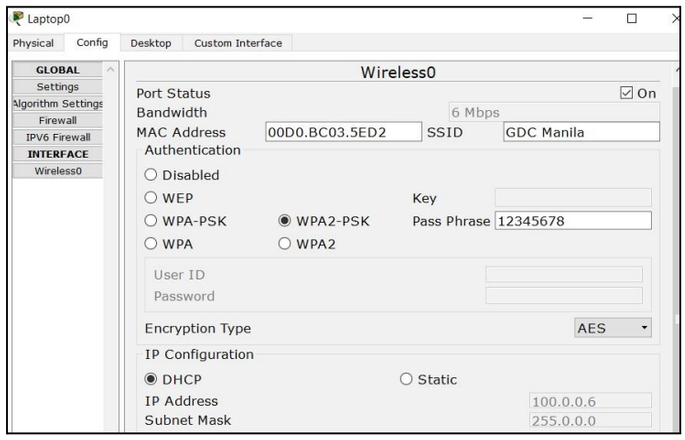
Manila Campus First Floor Communications Room

- The communication room consists of the Manila Campus router, main campus switch, and servers.
- Starting with the Manila Campus router, has a similar configuration to the Brighton router, with its unique IP addresses (100.0.0.1 & 192.168.1.1). This router is the only main router for the whole campus since the university is moderate size with three floors, thus can manage the routing requirements for the entire campus. Additionally, it is more simple to configure and troubleshoot in case of maintenance. This approach is more cost-effective as it requires fewer expenses and resources. I believe that this would be sufficient to accommodate the network needs of the campus.
- I also included in my network plan one of the servers that are essential for the university. This includes DHCP server, Web server, and DNS server. Although one server is capable of handling multiple services, I still provided each service one server just to show the servers available for the university and I could assign IP addresses for each. Multiple server hardware can provide redundancy in case one of the services fails, thus wouldn't be a problem for the overall system. This makes it also scalable especially when

  the number of users in the university increases, the servers can still independently distribute the services based on demands.
- I configured the three servers for them to be utilized by the devices with their respective IP addresses. I created an example of a web server/HTTP and DNS for the university website. The following pictures show how the DNS and web server are applied.



- All of the computers and laptops are successfully connected to the DNS and Web servers. While the computers employ static IP addresses, the laptops utilize dynamic IP addresses assigned by the DHCP server.



- I added an employee laptop to demonstrate that the right side access point is working on the first floor.

Computer Labs

- The computer lab can accommodate 20 computers, but initially, I installed 10. I provided one switch for each computer lab so that all of the computers could connect. Since each switch contains 24 ports, this ensures that it is future-proof when 10 more other computers are connected to the network.
- I added two laptops to demonstrate that the left side access point is working on the first floor.

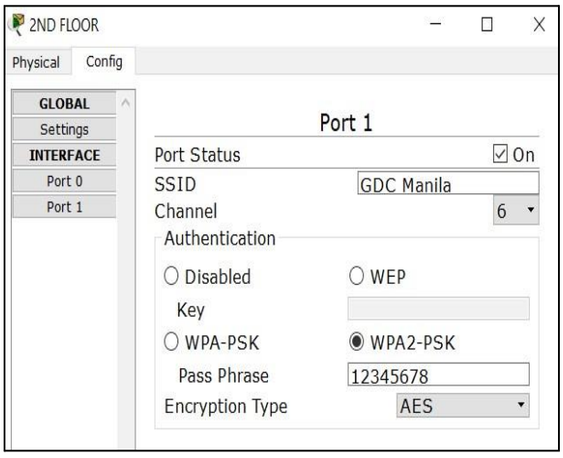
Second Floor Library

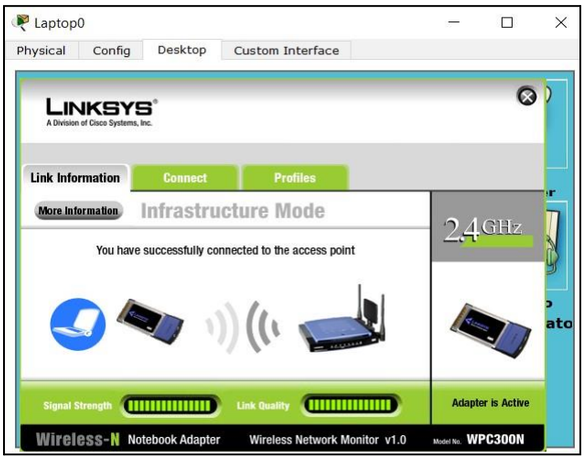
- The library contains 20 computers and one switch with 24 ports will be sufficient for it to accommodate. **Staff Offices**

- The staff office devices are connected to one switch since there are only 5 devices for each room and 2 ports allocated for the access point.
- For the IP phones, I inserted a power adapter module for it to be connected to the network.

Lecture Rooms

- I made the students' laptops connect wirelessly with the Linksys-WPC300N module and connect to port 1 of access points near their location. For the teachers' laptops, I provided a cable since it's a common practice in real-life scenarios and the connection will be more stable and secured.
- I configured the port 1 interface of all of the access points on each floor so that laptops can connect wirelessly. This is done by assigning an SSID and WPA2 with a Pass key for security purposes. The laptops are successfully connected to the network and therefore can access WIFI too.





- I provided another switch for lecture rooms to facilitate the connection of teachers' laptops. Since the staff offices switch is nearly occupied, there will be a necessity to install another switch to cater to lecture room laptops and new networked devices in the future.

Third Floor Staff Offices and Lecture Rooms/Hall

- The laptops, IP phones, and access points have a similar configuration to those devices on the second floor. This is to ensure consistency and same user experience across the areas.
- There is also one switch provided for the staff office and lecture rooms for the same reason on the second floor.

Periodic Network Maintenance Plan

A periodic Network Maintenance Plan is essential to prevent unexpected expenses from hardware or network failures, continuous operations, minimal downtime, and protection against potential threats.

NETWORK MAINTENANCE CHECKLIST	
DAILY	
☐	Check network connection and speed, and troubleshoot if necessary.
☐	Review device logs (router, switches, servers, computers).
☐	Observe the data flow or network traffic.

☐ Ensure that the lecture room and hall laptops are connected.
☐ Regularly backup crucial data in case of unexpected outages or maintenance.
WEEKLY
☐ Ensure that the DHCP, DNS, and Web servers are functioning seamlessly.
☐ Verify the connection of the access points and performance must be at an optimal state.
☐ Ensure that the network hardwares are clean. Cables must always be organised and tidy.
MONTHLY
☐ Inspect hardware devices and cables for repairs or any damages.
☐ Keep and update network documentation where important maintenance details are recorded.
☐ Update antivirus software on servers and computers on the campus.
IF NECESSARY
☐ Reconfiguration, replacement, or upgrade of network devices.
☐ Replace power adapters of IP phones.
☐ Ensure that security updates are applied to all networking devices.
☐ Keep track of the faculty that have access to important files and those who are not.
☐ Delete old files of students (computer labs and library), to free up storage space.

PDU Test Results

A PDU test or Power Distribution Unit is a device that is responsible for distributing electric power to networking devices. In simulation of the Cisco Packet Tracer, provides insights into the power consumption and behavior of devices under a certain network topology. PDU Tests may be either failed or successful. A successful test means it is functioning correctly while failed suggests there are disruptions or incorrect connections. This allows users to troubleshoot and identify issues by themselves in their network plan (TestGuy Electrical Testing Network, 2019)

First Floor Communication Room and Brighton Campus

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time (sec)	Periodic	Num	Edit	Delete
	Successful	BRIGHTON CAMPUS	MANILA CAMPUS	ICMP		0.000	N	0	(edit)	(delete)
	Successful	MANILA CAMPUS	BRIGHTON CAMPUS	ICMP		0.000	N	1	(edit)	(delete)
	Successful	BRIGHTON CAMPUS	PC39	ICMP		0.000	N	10	(edit)	(delete)
	Successful	WEB SERVER	PC39	ICMP		0.000	N	11	(edit)	(delete)
	Successful	DNS SERVER	PC14	ICMP		0.000	N	12	(edit)	(delete)
	Successful	DHCP SERVER	PC11	ICMP		0.000	N	13	(edit)	(delete)
	Successful	PC69	DNS SERVER	ICMP		0.000	N	14	(edit)	(delete)
	Successful	BRIGHTON CAMPUS	PC85	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC85	BRIGHTON CAMPUS	ICMP		0.000	N	3	(edit)	(delete)
	Successful	MANILA CAMPUS	DHCP SERVER	ICMP		0.000	N	4	(edit)	(delete)
	Successful	MANILA CAMPUS	WEB SERVER	ICMP		0.000	N	5	(edit)	(delete)
	Successful	MANILA CAMPUS	DNS SERVER	ICMP		0.000	N	6	(edit)	(delete)
	Successful	PC5	MANILA CAMPUS	ICMP		0.000	N	7	(edit)	(delete)
	Successful	MANILA CAMPUS	PC5	ICMP		0.000	N	8	(edit)	(delete)
	Successful	MANILA CAMPUS	PC84	ICMP		0.000	N	9	(edit)	(delete)

Computer Labs

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time (sec)	Periodic	Num	Edit	Delete
	Successful	CL1	CL2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	CL1	CL3	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC27	PC23	ICMP		0.000	N	10	(edit)	(delete)
	Successful	CL4	CL3	ICMP		0.000	N	2	(edit)	(delete)
	Successful	CL2	CL1	ICMP		0.000	N	3	(edit)	(delete)
	Successful	CL3	CL2	ICMP		0.000	N	4	(edit)	(delete)
	Successful	CL2	CL4	ICMP		0.000	N	5	(edit)	(delete)
	Successful	PC6	PC11	ICMP		0.000	N	6	(edit)	(delete)
	Successful	PC22	PC16	ICMP		0.000	N	7	(edit)	(delete)
	Successful	PC39	PC34	ICMP		0.000	N	8	(edit)	(delete)
	Successful	PC9	PC4	ICMP		0.000	N	9	(edit)	(delete)

Second Floor

Library

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time (sec)	Periodic	Num	Edit	Delete
	Successful	LIB	DHCP SERVER	ICMP		0.000	N	9	(edit)	(delete)
	Successful	LIB	DNS SERVER	ICMP		0.000	N	8	(edit)	(delete)
	Successful	LIB	LR	ICMP		0.000	N	1	(edit)	(delete)
	Successful	LIB	MANILA CAMPUS	ICMP		0.000	N	10	(edit)	(delete)
	Successful	LIB	PC49	ICMP		0.000	N	0	(edit)	(delete)
	Successful	LIB	SO	ICMP		0.000	N	2	(edit)	(delete)
	Successful	LIB	WEB SERVER	ICMP		0.000	N	7	(edit)	(delete)
	Successful	PC46	PC41	ICMP		0.000	N	5	(edit)	(delete)
	Successful	PC48	PC57	ICMP		0.000	N	3	(edit)	(delete)
	Successful	PC50	PC51	ICMP		0.000	N	4	(edit)	(delete)
	Successful	PC59	PC57	ICMP		0.000	N	6	(edit)	(delete)

Staff Offices

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time (sec)	Periodic	Num	Edit	Delete
	Successful	LR	PC64	ICMP		0.000	N	10	(edit)	(delete)
	Successful	PC60	PC62	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC63	PC61	ICMP		0.000	N	3	(edit)	(delete)
	Successful	PC63	PC64	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC66	SO	ICMP		0.000	N	5	(edit)	(delete)
	Successful	PC67	PC70	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC67	PC82	ICMP		0.000	N	12	(edit)	(delete)
	Successful	PC68	WEB SERVER	ICMP		0.000	N	7	(edit)	(delete)
	Successful	PC69	DHCP SERVER	ICMP		0.000	N	9	(edit)	(delete)
	Successful	PC71	DNS SERVER	ICMP		0.000	N	8	(edit)	(delete)
	Successful	PC72	PC60	ICMP		0.000	N	11	(edit)	(delete)
	Successful	SO	MANILA CAMPUS	ICMP		0.000	N	6	(edit)	(delete)
	Successful	SO	PC66	ICMP		0.000	N	4	(edit)	(delete)

Lecture Rooms

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time (sec)	Periodic	Num	Edit	Delete
	Successful	student laptop 2	teacher laptop	ICMP		0.000	N	0	(edit)	(delete)
	Successful	teacher laptop	student laptop 2	ICMP		0.000	N	1	(edit)	(delete)
	Successful	Laptop3	Laptop14	ICMP		0.000	N	10	(edit)	(delete)
	Successful	Laptop2	Laptop13	ICMP		0.000	N	11	(edit)	(delete)
	Successful	Laptop1	Laptop12	ICMP		0.000	N	2	(edit)	(delete)
	Successful	Laptop2	Laptop13	ICMP		0.000	N	3	(edit)	(delete)
	Successful	Laptop13	teacher laptop	ICMP		0.000	N	4	(edit)	(delete)
	Successful	student laptop 2	Laptop12	ICMP		0.000	N	5	(edit)	(delete)
	Successful	student laptop 2	Laptop3	ICMP		0.000	N	6	(edit)	(delete)
	Successful	Laptop14	teacher laptop	ICMP		0.000	N	7	(edit)	(delete)
	Successful	Laptop4	Laptop15	ICMP		0.000	N	8	(edit)	(delete)
	Successful	Laptop15	Laptop4	ICMP		0.000	N	9	(edit)	(delete)

Third Floor

Lecture Rooms and Hall

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time (sec)	Periodic	Num	Edit	Delete
	Successful	student laptop	teacher laptop 2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	teacher laptop 2	student laptop	ICMP		0.000	N	1	(edit)	(delete)
	Successful	Laptop7	teacher laptop 2	ICMP		0.000	N	10	(edit)	(delete)
	Successful	Laptop10	student laptop	ICMP		0.000	N	11	(edit)	(delete)
	Successful	Laptop6	Laptop5	ICMP		0.000	N	2	(edit)	(delete)
	Successful	Laptop7	Laptop9	ICMP		0.000	N	3	(edit)	(delete)
	Successful	Laptop10	Laptop8	ICMP		0.000	N	4	(edit)	(delete)
	Successful	Laptop8	Laptop10	ICMP		0.000	N	5	(edit)	(delete)
	Successful	Laptop5	Laptop7	ICMP		0.000	N	6	(edit)	(delete)
	Successful	teacher laptop 2	Laptop6	ICMP		0.000	N	7	(edit)	(delete)
	Successful	student laptop	Laptop5	ICMP		0.000	N	8	(edit)	(delete)
	Successful	Laptop5	student laptop	ICMP		0.000	N	9	(edit)	(delete)

Staff Offices

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time (sec)	Periodic	Num
	Successful	PC73	WEB SERVER	ICMP		0.000	N	9
	Successful	PC75	PC60	ICMP		0.000	N	0
	Successful	PC76	MANILA CAMPUS	ICMP		0.000	N	6
	Successful	PC77	DNS SERVER	ICMP		0.000	N	8
	Successful	PC77	PC74	ICMP		0.000	N	1
	Successful	PC78	PC76	ICMP		0.000	N	2
	Successful	PC79	DHCP SERVER	ICMP		0.000	N	7
	Successful	PC79	PC66	ICMP		0.000	N	4
	Successful	PC83	PC79	ICMP		0.000	N	10
	Successful	PC83	PC82	ICMP		0.000	N	3
	Successful	SO	PC73	ICMP		0.000	N	5

User Feedback Analysis

Understanding Global Dimensions College's requirements and goals is crucial to determine the network's scale, capacity, and performance needs. The rough plan from the onset, meets some of the requirements. Functionality will be a big factor in the implementation part. Further considerations in the network design will be the implementation of QoS, likewise the scalability and flexibility are paramount, ensuring the network can adapt to evolving demands and technological advancements. Ensure to identify the categories of your LAN cables as well.

According to the user feedback, the things I need to consider in my network plan for Global Dimensions College are the network's scale, capacity, and performance needs. It is said that my initial network plan meets the needs of the GDC requirements, and functionality can be enhanced through the implementation of QoS. The feedback also suggests considering the scalability and flexibility of the plan in order to adapt to future technological advancements and identify the categories of LAN cables that will be used for the network.

In terms of scale and capacity, the network plan I created has sufficient switches and access points for each floor that can accommodate the devices for each room, may it be wired or wireless. On the other hand, the performance can be measured by the bandwidth, network traffic, and testing.

Based on my design, I think the performance is satisfactory at an average level, but can be greatly improved by implementing QoS, as mentioned in the feedback. Quality of Server or QoS is a network technology that controls the network traffic to ensure that high-performance can be run under limited capacity. It prioritizes critical devices, and applications in high-demand periods (Fortinet, 2023) In short, QoS uses techniques that optimize the overall performance of the network. A key component of QoS is the bandwidth and traffic. I enhanced my bandwidth by using wired connections for computers and laptops, which improves the delivery of data from computers.

Before implementing QoS, I should analyse which rooms are critical to prioritizing and would require a lot of bandwidth. First, I highly prioritize the communication rooms as this is where the main campus switch, router, WAN link, and servers are located. This is where the overall communication between floors begins and it is expectedly demanding, as there will be loads of network traffic. The access points and wireless devices must be highly prioritized too since numerous cellphones and laptops are wirelessly connected to access points, wherein there's an influx frequency of simultaneous connections and high-demanding applications.

Next, it would be the staff offices since applications related to emailing or any administrative tasks are critical operations for the university.

Lecture rooms must be prioritized since video presentations, online meetings, and websites are frequently encountered applications in this room, prompting the need to prioritize traffic. Computer labs and library are time to time used by students, but opening up different software applications, websites, and hands-on activities can build up network traffic. Implementing QoS in critical areas will further provide the necessary bandwidth.

It is also mentioned in the feedback that identifying categories of my LAN cables is required. The cable that will be used for the connection between the Manila campus and Brighton campus is fiber optic cable since it is ideal for very long-distance transmission, which offers high bandwidth and low latency. Although this type of cable can be quite expensive, Single-Mode Optical Fiber can be used since according to GeeksforGeeks (2022), it is cheaper than multimode cables due to their higher production. Fiber optic cable is also used for internet connectivity from the UK to the Philippines

Meanwhile, Twisted-pair cables or ethernet cables will be used for computers, servers, access points, and between switches for connection. Unshielded Twisted Pair Cable (UTP) is known for its versatility and popularity among LAN networks (GeeksforGeeks, 2021). Cat5e category applies to my network plan since it is ideal for Ethernet, and Fast Ethernet connections that support up to 1 GBPS data which is sufficient for the design (Admin, 2022). It is also generally cost-effective and practical compared to STP.

Conclusion

The chosen topology is a hybrid of star and tree topology. Star topology is primarily evident in rooms wherein devices are directly connected to switches, while tree topology can be seen in the overall structure of the network plan, as all of the connections are directed to the router. (Cisco, 2020)

The design I created prioritizes its scalability and connectivity since there are plenty of available ports in the switches for new devices. Moreover, the expenses are reduced because of a single router for the university.

Redundancy and reliability are considered by deploying servers and access points. The wireless connectivity I established can accommodate 500 devices, making it flexible for simultaneous connecting and disconnecting devices on the campus. Web and DNS server for web pages, domain names, email, and academic resources, while DHCP server for wireless laptops and cellphones connected to the network.

The multilayer switch or the main switch for the Manila Campus has a feature, Power over Ethernet (PoE), that is beneficial for devices that use ethernet cables like switches, IP phones, computers, and access points. This switch also allows the IT administrators to troubleshoot multiple switches at the same time. Although this switch comes at a higher cost than regular switches, it guarantees network performance and reliability when troubleshooting (Cisco, 2019).

In terms of cables, the fiber optic cable is applicable for long-distance connection, while maintaining high-speed data between the Manila campus and Bright campus. Single-mode optical fiber can be used to save up expenses. For the LAN network, UTP Cat5e is ideal for Fast Ethernet connections that can support up to 1000Mbps of data, which is sufficient for the network while offering a good amount of price.

The communication between the Brighton and Manila campuses has successfully managed WAN infrastructure through point-to-point serial link and static routing. (Community College WAN Deployment Guide, n.d.). Thus, allows internet access to the Manila campus. My network plan reflects the current requirements of GDC, and more improvements can be made for future technological needs.

Recommendations

My network design can be further improved by utilizing VLAN so that different departments in GDC can be segmented into different networks, especially if the university increases the number of departments. Implementing firewalls is also essential for the network security of the school since network worms or viruses can potentially intrude. In my network design, I implemented static routing, dynamic addressing, and WPA2 on access points that contribute to the overall security measures.

GDC should also have a skilled IT management team that can troubleshoot and regularly monitor network usage, ensuring that web pages and communications are at optimal performance. They are also responsible for reconfiguring the network if there are necessary changes. The IT department can use network systems like backup and load balancing. The periodic maintenance plan must be followed.

In terms of hardware, switches and routers will be effective for redundancy in case of failures and scalability requirements. In the future, upgrading and adding access points and switches will be necessary when users increase, like occurrences of school events and activities. As mentioned, implementation of QoS is also advised for my network plan, which I will integrate in the future.

The implementation of QoS will help control the network traffic especially when the users are opening high-data applications. This is highly recommended for the network plan since it will speed up and prioritize the performance of the network, which would also enhance longevity and efficiency.

Strategic cable choices like fiber optics for campus connection and UTP Cat5e for internal LAN networks will ensure efficiency and practicality. Fiber optics and UTP Cat5e are the cables I recommended for the network plan.

For the network plan to be future-proof and prevent obsolescence after 2-3 years, investing in upgradable components of the hardware is suitable since there won't be a need to replace the entire device. This applies to switches, routers, and computers, which can save expenses and effort for the IT team. As I mentioned, regularly monitoring the network is important in order to address problems immediately without prolonging it and corrupting the network, which could affect servers. That's why the need for advanced security technologies must be implemented like firewalls or detection systems.

Moreover, I recommend switches with 48 ports so that it would accommodate more devices and wouldn't be a workload to set up too many switches in rooms. Documentation is also essential for keeping a record of network device configuration, IP addresses, and troubleshooting. These are the recommendations that will improve the network design and utilization.

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VLSM Table

Manila Campus	100.25.0.0/10
Brighton Campus	90.100.10.20/8

MANILA CAMPUS

First Floor

Subnet Name	Needed Size	Allocated Size	Network Address	CIDR	Subnet Mask	Wildcard Mask	Usable Range	Broadcast Address
Internet	500	510	100.0.0.0	/23	255.255.254.0	0.0.1.255	100.0.0.1 - 100.0.1.254	100.0.1.255
Computer Lab 1	20	30	100.0.2.0	/27	255.255.255.224	0.0.0.31	100.0.2.1 - 100.0.2.30	100.0.2.31
Computer Lab 2	20	30	100.0.2.32	/27	255.255.255.224	0.0.0.31	100.0.2.33 - 100.0.2.62	100.0.2.63
Computer Lab 3	20	30	100.0.2.64	/27	255.255.255.224	0.0.0.31	100.0.2.65 - 100.0.2.94	100.0.2.95
Computer Lab 4	20	30	100.0.2.96	/27	255.255.255.224	0.0.0.31	100.0.2.97 - 100.0.2.126	100.0.2.127
Second Floor								
Library	20	30	100.0.2.128	/27	255.255.255.224	0.0.0.31	100.0.2.129 - 100.0.2.158	100.0.2.159
Staff Office 1	5	6	100.0.2.160	/29	255.255.255.248	0.0.0.7	100.0.2.161 - 100.0.2.166	100.0.2.167
Staff Office 2	5	6	100.0.2.168	/29	255.255.255.248	0.0.0.7	100.0.2.169 - 100.0.2.174	100.0.2.175
Staff Office 3	5	6	100.0.2.220	/29	255.255.255.248	0.0.0.7	100.0.2.177 - 100.0.2.182	100.0.2.183
Staff Office 4	5	6	100.0.2.224	/29	255.255.255.248	0.0.0.7	100.0.2.185 - 100.0.2.190	100.0.2.181
Lecture Room 1	1	2	100.0.2.228	/30	255.255.255.252	0.0.0.3	100.0.2.221 - 100.0.2.222	100.0.2.223
Lecture Room 2	1	2	100.0.2.232	/30	255.255.255.252	0.0.0.3	100.0.2.225 - 100.0.2.226	100.0.2.227
Lecture Room 3	1	2	100.0.2.236	/30	255.255.255.252	0.0.0.3	100.0.2.229 - 100.0.2.230	100.0.2.231
Lecture Room 4	1	2	100.0.2.176	/30	255.255.255.252	0.0.0.3	100.0.2.233 - 100.0.2.234	100.0.2.235
Lecture Room 5	1	2	100.0.2.184	/30	255.255.255.252	0.0.0.3	100.0.2.237 - 100.0.2.238	100.0.2.239
Third Floor								
Staff Office 1	5	6	100.0.2.192	/29	255.255.255.248	0.0.0.7	100.0.2.193 - 100.0.2.198	100.0.2.199
Staff Office 2	5	6	100.0.2.200	/29	255.255.255.248	0.0.0.7	100.0.2.201 - 100.0.2.206	100.0.2.207
Staff Office 3	5	6	100.0.2.240	/29	255.255.255.248	0.0.0.7	100.0.2.212 - 100.0.2.217	100.0.2.218
Staff Office 4	5	6	100.0.2.244	/29	255.255.255.248	0.0.0.7	100.0.2.220 - 100.0.2.225	100.0.2.226
Lecture Room 1	1	2	100.0.2.248	/30	255.255.255.252	0.0.0.3	100.0.2.241 - 100.0.2.242	100.0.2.243
Lecture Room 2	1	2	100.0.2.252	/30	255.255.255.252	0.0.0.3	100.0.2.245 - 100.0.2.246	100.0.2.247
Lecture Room 3	1	2	100.0.3.0	/30	255.255.255.252	0.0.0.3	100.0.2.249 - 100.0.2.250	100.0.2.251
Lecture Hall	1	2	100.0.3.32	/30	255.255.255.252	0.0.0.3	100.0.2.253 - 100.0.2.254	100.0.2.255

Brighton Campus

Subnet Name	Needed Size	Allocated Size	Network Address	CIDR	Subnet Mask	Wildcard Mask	Usable Range	Broadcast Address
Brighton Campus/Testing Room	2	2	90.0.0.0	/30	255.255.255.252	0.0.0.3	90.0.0.1 - 90.0.0.2	90.0.0.3

PDU Test: Actual vs Expected Result

Source	Destination	Expected	Actual
MANILA CAMPUS ROUTER	BRIGHTON CAMPUS ROUTER	Successful	Successful
BRIGHTON CAMPUS ROUTER	MANILA CAMPUS ROUTER	Successful	Successful
BRIGHTON CAMPUS ROUTER	PC85	Successful	Successful
PC85	BRIGHTON CAMPUS ROUTER	Successful	Successful
PC84	PC85	Successful	Successful
PC85	PC84	Successful	Successful
CL1	CL2	Successful	Successful
CL2	CL1	Successful	Successful
CL3	CL4	Successful	Successful
CL4	CL3	Successful	Successful
WEB SERVER	MANILA CAMPUS ROUTER	Successful	Successful
DNS SERVER	MANILA CAMPUS ROUTER	Successful	Successful
DHCP SERVER	MANILA CAMPUS ROUTER	Successful	Successful
LIB	PC49	Successful	Successful
PC49	LIB	Successful	Successful
PC42	PC50	Successful	Successful
student laptop 2	teacher laptop	Successful	Successful
teacher laptop	student laptop 2	Successful	Successful
Laptop2	Laptop14	Successful	Successful

Laptop3	Laptop15	Successful	Successful
PC60	PC61	Successful	Successful
PC76	PC66	Successful	Successful
PC61	PC60	Successful	Successful
student laptop	teacher laptop 2	Successful	Successful
teacher laptop 2	student laptop	Successful	Successful
PC75	PC80	Successful	Successful
PC80	PC75	Successful	Successful
Laptop8	Laptop10	Successful	Successful
Laptop6	Laptop7	Successful	Successful
BRIGHTON CAMPUS ROUTER	PC60	Successful	Successful